

MATHEMATICS**Maximum Marks: 80****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional fifteen minutes for only reading the paper.
2. You must NOT start writing during reading time.
3. The question paper has 9 printed pages.
4. It consists of twenty questions and four sections: Sections A, B, C and D.
5. All questions are compulsory.
6. Section A consists of very short answer questions of 1 mark each.
7. Section B consists of short answer questions of 2 marks each.
8. Section C consists of moderately long answer questions of 3 marks each.
9. Section D consists of long answer questions of 5 marks each.
10. Internal choices have been provided in three questions, each in Sections B, C and D.
11. While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
12. The intended marks for questions or parts of questions are given in the brackets [].
13. All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
14. Mathematical tables and graph papers are provided.

Instruction to Supervising Examiner

Kindly read aloud the instructions given above to all the candidates present in the examination hall.

SECTION A – 20 MARKS**Question 1**

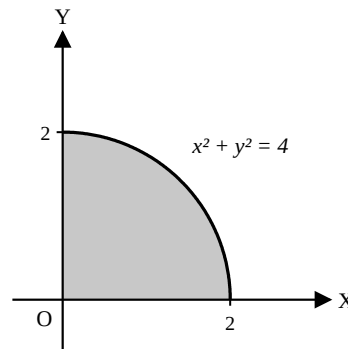
In subparts (i) to (xvii) choose the correct options and in subparts (xviii) to (xx), answer the questions as instructed.

- (i) R is the relation on $\mathbb{Z}$ defined by aRb if and only if $a - b$ is divisible by 3. The number of distinct equivalence classes of R is: (*Understand*) [1]
- (a) 1
 - (b) 2
 - (c) 3
 - (d) infinitely many
- (ii) The value of $\tan^{-1}\sqrt{3} - \sec^{-1}(-2)$ is: (*Understand*) [1]
- (a) π
 - (b) $-\pi/3$
 - (c) $\pi/3$
 - (d) $2\pi/3$

- (iii) If the area of the triangle with vertices $(-2, 0)$, $(0, 4)$ and $(0, k)$ is 4 square units, then: (*Apply*) [1]
- (a) $k = 0$ or 8
 - (b) $k = 0$ or -8
 - (c) $k = 4$ only
 - (d) $k = 8$ only
- (iv) A particle moves along the curve $y^2 = 8x$. The point on the curve at which the x-coordinate is changing twice as fast as the y-coordinate is: (*Apply*) [1]
- (a) $(2, 4)$
 - (b) $(8, 8)$
 - (c) $(8, -8)$
 - (d) $(4, 4\sqrt{2})$
- (v) The height (in m) of a ball thrown vertically upwards is $h(t) = 20t - 5t^2$ after t seconds. [1]
Statement I: The ball is rising during $0 < t < 2$.
Statement II: The maximum height reached by the ball is 40 m.
Which of the following is correct? (*Understand*)
- (a) Statement I is true and Statement II is false.
 - (b) Statement I is false and Statement II is true.
 - (c) Both the statements are true.
 - (d) Both the statements are false.
- (vi) The curves $ay + x^2 = 7$ and $x^3 = y$ cut each other orthogonally at $(1, 1)$. The value of a is: [1]
(*Apply*)
- (a) 1
 - (b) 0
 - (c) -6
 - (d) 6
- (vii) The function $f(x) = x + 1/x$, $x \neq 0$, has: (*Analyse*) [1]
- (a) a local maximum value 2 at $x = 1$ and a local minimum value -2 at $x = -1$
 - (b) a local maximum value -2 at $x = -1$ and a local minimum value 2 at $x = 1$
 - (c) only a local minimum, of value 2, at $x = 1$
 - (d) neither a local maximum nor a local minimum

(viii) The expression for the area of the shaded region shown below is: (*Apply*)

[1]



(a) $\int_0^2 (4 - x^2) dx$

(b) $\int_0^2 \sqrt{4 - x^2} dx$

(c) $\int_{-2}^2 \sqrt{4 - x^2} dx$

(d) $\int_0^4 \sqrt{4 - x^2} dx$

(ix) Shown below is a solution of $\int \tan x dx$:

[1]

Step 1 $\int \tan x dx = \int \frac{\sin x}{\cos x} dx$

Step 2 Put $\cos x = t$, so $-\sin x dx = dt$. Then the integral $= \int \frac{dt}{t}$

Step 3 $= \log|t| + C = \log|\cos x| + C$

In which step is there an error (if any)? (*Understand*)

(a) Step 1

(b) Step 2

(c) Step 3

(d) No error

(x) On evaluating $\int \log x dx$, Priya's answer was $x \log x - x + c$ and Rohan's answer was $x \log(x/e) + c$. Whose answer is correct? (*Understand*)

[1]

(a) Only Priya's answer is correct.

(b) Only Rohan's answer is correct.

(c) Both of them are correct.

(d) Neither of them is correct.

(xi) If $|\vec{a} \times \vec{b}| = 4$ and $|\vec{a} \cdot \vec{b}| = 2$, then $|\vec{a}|^2 |\vec{b}|^2$ is equal to: (*Apply*)

[1]

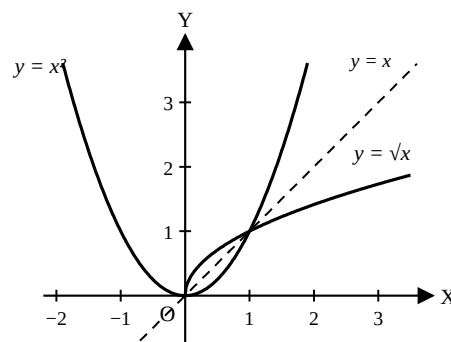
(a) 6

(b) 8

(c) 20

(d) 2

- (xii) The angle between the lines $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $\frac{x-5}{4} = \frac{y-2}{1} = \frac{z-3}{8}$ is: (Apply) [1]
- (a) $\cos^{-1}(2/3)$
 (b) $\cos^{-1}(1/3)$
 (c) $\pi/3$
 (d) $\pi/2$
- (xiii) The corner points of the feasible region of an LPP are (0, 0), (0, 40), (20, 40), (60, 20) and (60, 0). The maximum value of $Z = 4x + 3y$ is: (Apply) [1]
- (a) 240
 (b) 300
 (c) 200
 (d) 120
- (xiv) Statement I: The region given by $x \geq 0$, $y \geq 0$ and $x + y \geq 4$ is unbounded. [1]
 Statement II: On the region in Statement I, $Z = 2x + 3y$ has no minimum value.
 Which of the statements is correct? (Analyse)
- (a) Statement I is true and Statement II is false.
 (b) Statement I is false and Statement II is true.
 (c) Both the statements are true.
 (d) Both the statements are false.
- (xv) A speaks the truth in 60% of cases and B in 90% of cases. The probability that they contradict each other in stating the same fact is: (Apply) [1]
- (a) 0.42
 (b) 0.54
 (c) 0.46
 (d) 0.58
- (xvi) Observe the graphs of $y = x^2$ ($x \in \mathbb{R}$) and $y = \sqrt{x}$ given below. [1]
 Assertion: The two curves are the graphs of $f(x) = x^2$, $x \in \mathbb{R}$, and its inverse f^{-1} .
 Reason: The graph of f^{-1} is the reflection of the graph of f in the line $y = x$.
 Which one of the following is correct? (Analyse)



- (a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
 (c) Assertion is true and Reason is false.
 (d) Assertion is false and Reason is true.

(xvii) Assertion: $\int_{-\pi/2}^{\pi/2} \cos x \, dx = 2$. [1]

Reason: If f is a continuous even function, then $\int_{-a}^a f(x) \, dx = 2 \int_0^a f(x) \, dx$.

Which one of the following is correct? (*Understand*)

- (a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
- (b) Both Assertion and Reason are true, but Reason is not the correct explanation of Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(xviii) If A is a square matrix such that $A^2 = A$, simplify $(I + A)^2 - 3A$. (*Apply*) [1]

(xix) Form the differential equation of the family of curves $y = ax^2$, where a is an arbitrary constant. (*Apply*) [1]

(xx) If $P(A) = 0.4$, $P(B | A) = 0.5$ and $P(A | B) = 0.25$, find $P(B)$. (*Apply*) [1]

SECTION B – 14 MARKS

Question 2

Find the angle between the planes $\vec{r} \cdot (2\hat{i} - \hat{j} + \hat{k}) = 6$ and $\vec{r} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 5$. (*Apply*) [2]

Question 3

Six boys and four girls sit in a row of ten chairs at random. Find the probability that no two girls sit next to each other. (*Evaluate*) [2]

Question 4

Solve the differential equation $(x^2 - y^2) \, dx + 2xy \, dy = 0$, given that $y = 1$ when $x = 1$. (*Apply*) [2]

Question 5

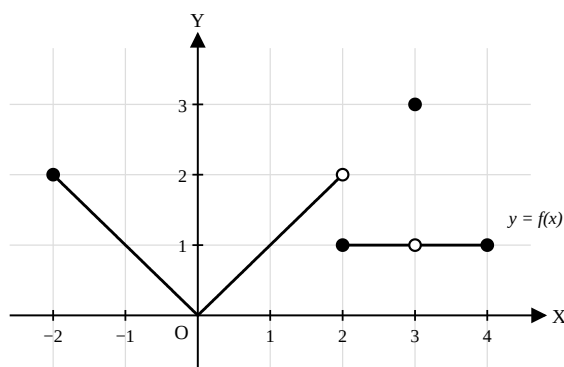
Find the values of a and b such that the function [2]

$$f(x) = \begin{cases} x^2 + 3x + a, & x \leq 1 \\ bx + 2, & x > 1 \end{cases}$$

is differentiable at $x = 1$. (*Understand*)

OR

The graph of $y = f(x)$, $-2 \leq x \leq 4$, is given below.



(a) State the points of discontinuity of f and whether each is removable. (*Understand*) [1]

- (b) State the points in $(-2, 4)$ where f is not differentiable. (*Understand*) [1]

Question 6

Solve for x : $\tan^{-1}\left(\frac{x-1}{x-2}\right) + \tan^{-1}\left(\frac{x+1}{x+2}\right) = \pi/4$. (*Apply*) [2]

OR

Find the real numbers x and a for which $ax + \sin^{-1}(x^2 - 4x + 5) + \cos^{-1}(x^2 - 4x + 5) = 0$. (*Apply*) [2]

Question 7

Find a vector of magnitude 9 which is perpendicular to both $\vec{a} = 4\hat{i} - \hat{j} + 3\hat{k}$ and $\vec{b} = -2\hat{i} + \hat{j} - 2\hat{k}$. (*Apply*) [2]

OR

Find λ if the scalar projection of $\vec{a} = \lambda\hat{i} + \hat{j} + 4\hat{k}$ on $\vec{b} = 2\hat{i} + 6\hat{j} + 3\hat{k}$ is 4 units. (*Apply*) [2]

Question 8

A hall is a cuboid 8 m long, 6 m wide and 5 m high. A corner O of the floor is taken as origin, with $\hat{i}, \hat{j}$ along the floor edges of lengths 8 m and 6 m and $\hat{k}$ vertically upwards. A lamp L hangs at the centre of the ceiling, joined to O by a straight wire.

- (i) Find the position vector of L and the length of the wire OL. (*Understand*) [1]
(ii) Find the angle which the wire makes with the vertical edge through O. (*Apply*) [1]

SECTION C – 21 MARKS

Question 9

Minimise $Z = 3x + 5y$ subject to $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0$, $y \geq 0$.

- (i) Draw the feasible region and find its corner points. (*Apply*) [1]
(ii) Evaluate Z at the corner points. (*Apply*) [1]
(iii) The region is unbounded. Justify that the smallest corner value is the minimum of Z . (*Evaluate*) [1]

Question 10

If $x = e^t$ and $y = f(x)$, prove that $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} = \frac{d^2y}{dt^2}$. (*Apply*) [3]

Question 11

A curve is given by $x^y = e^{x-y}$, $x > 0$.

- (i) Prove that $\frac{dy}{dx} = \frac{\log x}{(1 + \log x)^2}$. (*Apply*) [2]
(ii) Find the equation of the tangent to the curve at the point where $x = 1$. (*Apply*) [1]

Question 12

Find the equation of the plane passing through the points A(1, 1, 0), B(1, 2, 1) and C(-2, 2, -1). Also find its distance from the origin. (*Apply*) [3]

OR

A line L passes through (1, 2, -4) and is perpendicular to each of the lines

$$\frac{x-8}{3} = \frac{y+19}{-16} = \frac{z-10}{7} \quad \text{and} \quad \frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}.$$

- (a) Find direction ratios of L. (*Apply*) [1]
 (b) Write the equation of L in Cartesian and vector form. (*Understand*) [1]
 (c) Find the angle that L makes with the z-axis. (*Apply*) [1]

Question 13

Using properties of determinants, prove that [3]

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c).$$

(*Analyse*)

OR

An NGO buys three types of food packets P, Q, R for three villages. Village X needs 20, 15 and 10 packets; village Y needs 25, 10 and 5; village Z needs 10, 20 and 15. The packets cost ₹50, ₹80 and ₹100 each. Using matrix multiplication, find the cost for each village. If the NGO has a budget of ₹9,000, is it sufficient? (*Analyse*) [3]

Question 14

Let $I = \int_0^{\pi} \frac{x}{1 + \sin x} dx.$

- (i) Show that $I = \int_0^{\pi} \frac{\pi - x}{1 + \sin x} dx.$ (*Understand*) [1]
 (ii) Hence evaluate I. (*Apply*) [2]

Question 15

Using integration, find the area of the smaller region cut off from the circle $x^2 + y^2 = 4$ by the line $x + y = 2$. (*Analyse*) [3]

OR

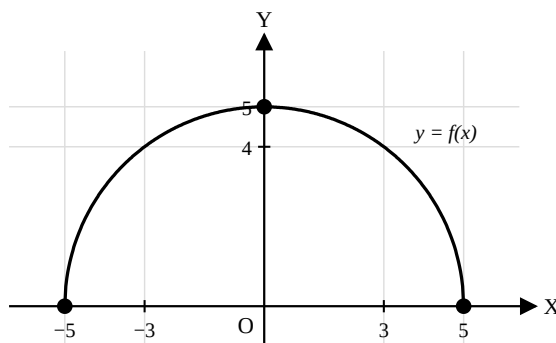
The tangent to the parabola $y = x^2$ at P(1, 1) meets the x-axis at Q.

- (a) Find the equation of the tangent and the point Q. (*Apply*) [1]
 (b) Using integration, find the area bounded by the parabola, the tangent and the x-axis. (*Apply*) [2]

SECTION D – 25 MARKS

Question 16

The cross-section of a tunnel is the curve $y = f(x) = \sqrt{25 - x^2}$ shown below.



- (i) Find the natural domain of f . (*Understand*) [1]
- (ii) Find the range of f , and hence the codomain for which f is onto. (*Understand*) [1]
- (iii) Is f one-one on $[-5, 5]$? Justify. (*Analyse*) [1]
- (iv) Let g be f restricted to $[0, 5]$, $g : [0, 5] \rightarrow [0, 5]$. Find $g^{-1}(x)$. (*Apply*) [1]
- (v) Find the point where the graph of g meets the line $y = x$. (*Analyse*) [1]

Question 17

The height (in tens of metres) of a trekking trail is $h(x) = x^4 - 8x^2 + 20$, where x (km) is the horizontal position measured from a base camp, positive to the east and negative to the west.

- (a) Find the intervals in which h is increasing and decreasing. (*Understand*) [1]
- (b) Find the local maximum and local minimum values of h . (*Understand*) [1]
- (c) Find the absolute maximum and minimum heights on the stretch $-1 \leq x \leq 3$. (*Analyse*) [2]
- (d) Find the equation of the tangent to $y = h(x)$ at $x = 1$. (*Apply*) [1]

OR

A window is a rectangle surmounted by a semicircle whose diameter is the top of the rectangle. The perimeter of the window is 10 m. Let r be the radius of the semicircle.

- (a) Express the area A of the window as a function of r . (*Apply*) [2]
- (b) Find dA/dr . (*Understand*) [1]
- (c) Find the dimensions for which the window admits the maximum light, and the maximum area. (*Evaluate*) [2]

Question 18

- (a) Prove that $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$. (*Understand*) [2]
- (b) Hence evaluate $\int e^x \frac{1 + \sin x}{1 + \cos x} dx$. (*Evaluate*) [3]

OR

- (a) Evaluate $\int \frac{1}{x^2 + 2x + 5} dx$. (*Understand*) [2]
- (b) Hence evaluate $\int \frac{3x + 1}{x^2 + 2x + 5} dx$. (*Evaluate*) [3]

Question 19

Box I contains two gold coins, Box II two silver coins and Box III one gold and one silver coin. A box is chosen at random and a coin is taken out of it.

- (a) Find the probability that the coin is gold. (*Apply*) [2]
- (b) The coin is gold. Find the probability that it came from Box I. (*Analyse*) [2]
- (c) A student argues that the other coin in the same box is gold with probability $1/2$. Evaluate the argument. (*Evaluate*) [1]

OR

In a game a player throws a die at most three times. He wins ₹5 if a number greater than 4 appears and then stops; otherwise he loses ₹1 and throws again. X is his net gain.

- (a) Write the possible values of X . (*Understand*) [1]
- (b) Find the probability distribution of X . (*Apply*) [2]
- (c) Find his expected gain. (*Evaluate*) [2]

Question 20

Ravi buys 3 kg apples, 2 kg oranges and 1 kg grapes for ₹510. Sita buys 2 kg apples, 1 kg oranges and 3 kg grapes for ₹530. Tanu buys 1 kg apples, 3 kg oranges and 2 kg grapes for ₹460. Taking the prices per kg as ₹ x , ₹ y and ₹ z , write the equations as $AX = B$ and solve them by the matrix method. (*Evaluate*) [5]

— — — —